AP PRECALCULUS

Unit 1 • Lesson 1.3 notes

Lesson 1.3 • Rates of Change in Linear and Quadratic Functions

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Recognize linear functions by their constant rate of change.
- Recognize quadratic functions by their constant second differences.
- Connect the sign of the leading coefficient of a quadratic to its concavity.

Vocabulary. linear function, slope, quadratic function, first differences, second differences, vertex.

1. Linear functions

A linear function $f(x) = mx + b$ has the same average rate of change, m, on every interval. In a table with equally spaced inputs, the outputs change by the same amount each step.

2. Quadratic functions

For $f(x) = ax^2 + bx + c$, the average rates of change over consecutive intervals of the same length change by a constant amount.

Key idea

With inputs equally spaced by h, the **second differences** of a quadratic are all equal to $2ah^2$. In particular, for $h = 1$ they equal $2a$.

Example 1 • Finding a quadratic from a table

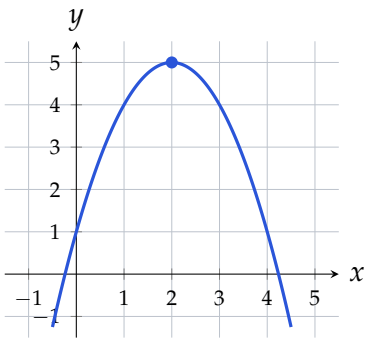
x	0	1	2	3	4
y	3	1	-3	-9	-17
first differences		-2	-4	-6	-8
second differences			-2	-2	-2

Solution. Constant second differences: quadratic, with $2a = -2$, so $a = -1$. The y-intercept is $c = 3$. Using $x = 1$: $-1 + b + 3 = 1$, so $b = -1$. The function is $y = -x^2 - x + 3$.

3. Concavity and the vertex

If $a > 0$, the rate of change is increasing: the parabola is concave up and has a minimum at the vertex. If $a < 0$, it is concave down with a maximum. At the vertex $x = -\frac{b}{2a}$ the rate of change changes sign.

Example 2 • Graph



Solution. The vertex is $(2, 5)$ and the graph passes through $(0, 1)$: $1 = a(0-2)^2 + 5$ gives $a = -1$, so $f(x) = -(x-2)^2 + 5$. Average rates of change: $[0,1]$: 3; $[1,2]$: 1; $[2,3]$: -1. They decrease by 2 each step: concave down.

Common mistakes

- Using second differences when the inputs are not equally spaced.
- Thinking that second differences equal a (they equal $2a$ when the step is 1).

Quick check

1. Outputs at $x = 0, 1, 2, 3$ are $2, 5, 10, 17$. Linear, quadratic, or neither?
2. A quadratic has second differences of 6 for inputs spaced by 1. What is a? Is it concave up or down?

Answers. 1. Differences $3, 5, 7$, second differences 2: quadratic ($y = x^2 + 2x + 2$). 2. $a = 3$; concave up.